



Motion Down the Incline

3.1 Program Outcomes (POs)

At the end of the term, the students must be able to:

- communicate effectively.
- design and conduct experiments, as well as to analyze and interpret data

3.2 Objectives of the Activity

At the end of the activity, the students must be able to:

- measure the acceleration of an object in an inclined plane.
- formulate a quantitative relationship between the acceleration and the angle of inclination.
- experimentally determine the magnitude of the acceleration due gravity.

3.3 Principle of the Activity

An object moving down a frictionless inclined plane is an example of uniformly accelerated motion. The object accelerates due to the action of the earth's gravity.

Ignoring friction, the acceleration component of the cart down the incline can be shown to be $a = g \sin\theta$, where θ is the angle of the incline above the horizontal.

If we know the height h and length L of the track then we know that $\sin\theta = h/L$

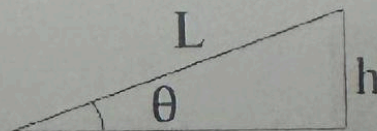


Figure 3.1. Relation of θ , L and h

Therefore, the acceleration of the object moving down a frictionless inclined plane is

$$a = g \left(\frac{h}{L} \right)$$

3.4 Materials/Equipment

Qty	Item	Group	Class
1	PASCO dynamics track with end stops	Measuring Tools	0
1	PASCO smart cart	Measuring Tools	0
1	Meter stick	Measuring Tools	0
1	PASCO Capstone software	Others	0
1	PASCO pivot clamp	Others	0
3	Mass bars	Others	

3.5 Procedure/s

3.5.1 Setting Up

1. Set up the track as shown in Figure 3.1 using the rod base and 45-cm rod. Note that the lower end of the track rests directly on the table, and that the track feet are NOT used in this experiment. The square nut on the Track Rod Clamp slides into the T-Slot on the track, and allows the track to be secured at various angles.

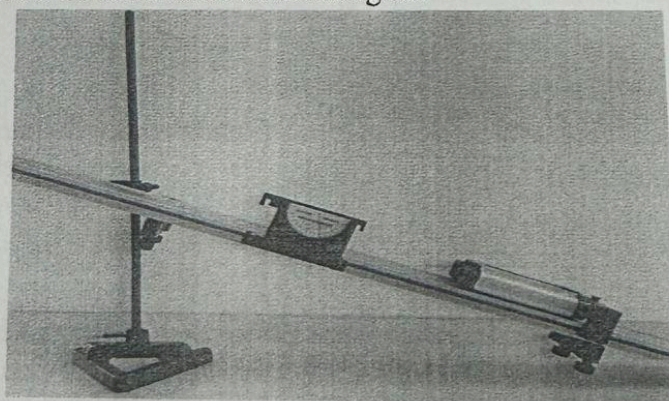
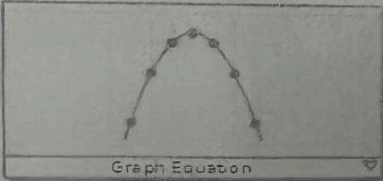


Figure 3.1 Set up of the inclined track

2. Install the fixed End Stop at the top and at the bottom of the incline.
3. Place the Smart Cart on the track so that its +x-direction, printed on top of the cart, is pointing down the track.

4. Turn on the Smart Cart. In PASCO Capstone, click  and connect it.

5. Click  or  to make a graph of velocity vs. time. Label the y-axis with velocity (m/s) and the x-axis with time (s).

3.5.2 Measuring Acceleration

1. Determine the length of the cart. Denote it as L . Take note of this length.
2. Adjust the height h of the upper end of the track to about 20 cm. The height should be measured to the underside of the track as shown in Figure 3.2.

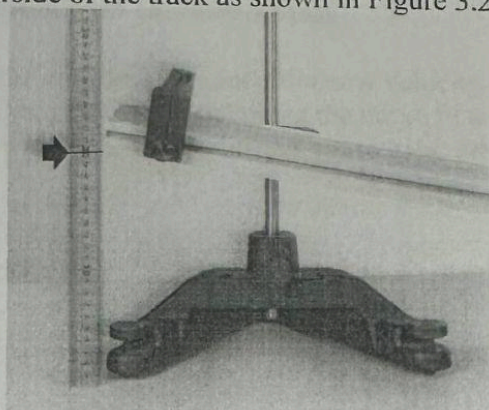


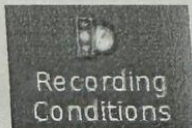
Figure 3.2. Measure the height of the elevation of the track

3. Calculate the theoretical value of the acceleration of the cart down the incline. Use the equation below and record the value in Table 3.6.1.

$$a_{theo} = g \left(\frac{h}{L} \right)$$

where g is the gravitational acceleration

$$g = 9.81 \text{ m/s}^2$$

4. In Capstone, before recording data on the velocity vs. time graph, click  to set the following Recording Conditions on the lower toolbar: a Start Condition based on the Position measurement, is above 0.10 m; and a Stop Condition based on the Position measurement, is above 0.70 m. These settings make it easier to get clean data because, after you click Record, actual data recording starts only after the cart has moved 10 cm, and automatically stops when it reaches 70 cm.
5. Hold the cart at rest at the highest point on the inclined track, making sure the +x-direction printed on top of the cart is pointing down the incline.

6. Click  to start recording data and release the cart. Repeat if needed until you get a nice velocity graph. You can delete unwanted runs using the Delete feature in the lower

toolbar,

Delete Last Run

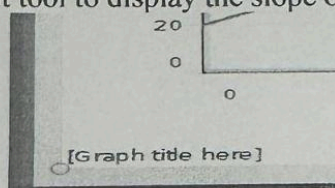
Data Summary

7. Click the Data Summary and right click on run 1 to rename it. Type '500g mass'.

- Click  to turn on the curve fit tool on the graph tool palette, and select a Linear fit. Determine the **slope** of the line. Denote this as the **experimental value of the acceleration** of the cart.

3.5.3 Measuring Acceleration with Increased Mass

- Add the mass bar to the cart, and then record a new velocity graph.
- Measure its new acceleration (the slope) using the curve fit tool and indicate it in the graph.
- Repeat the previous with different additional mass bars, 250g, 500g, 1750g and 1000g, respectively.
- Name the runs
- Turn on the multi-dataset tool  in the graph tool palette, and select all runs to display all the graphs.
- Use the curve fit tool to display the slope of the fit lines.



- Name the graph  Motion of the Cart of Different Masses.
- Use the Journal tool  to take a snapshot of this display.

3.5.4 Measuring Acceleration with Increased Height



- Click  create a new table with three columns. In the first column, create a user-entered data called **h** in units of m.
- In the second column, create a new calculation. In the calculation editor that appears on the table, enter: " h/L "= $[h(m)]/1.22$ with no units. (Note: This assumes your track length is $L = 1.22$ m, but you should change it to match your measured value of L here.)
- Click  to add a third column with user-entered data called Cart Acceleration in units of m/s^2 . On the unit, right click to show the command for superscript.
- In the first row of the table, use the data from your previous data run of the cart without the extra mass.
- Enter 0.20 in the **h** column. The " h/L " column should calculate automatically.
- In the Cart Acceleration column, enter the value of the slope of the velocity vs. time graph.
- Raise the height to 0.25 m, being careful to measure to the lower side of the track as shown in Figure 3.2.
- Record a new velocity vs. time graph, and measure the new acceleration. Record these values in the next row of the table.
- Repeat for track heights of 0.30 m and 0.35 m.
- Name the runs
- Click  to create a new page.

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12. Click  to create a graph with Cart Acceleration on the vertical axis and “h/L” on the horizontal axis. Turn on the curve fit tool, and select the Linear fit. Note that the slope is the theoretical value of the gravitational acceleration.
 13. Name the graph Acceleration Versus h/L.
 14. Use the Journal tool to take a snapshot of the display, indicating the graphs and the table generated.
 15. Compare the experimental and theoretical values of the gravitational acceleration.

$$\%error_g = \frac{|g_{theo} - g_{exp}|}{g_{theo}} \times 100$$

3.6 Data and Results

Attach all the tables and graphs generated in Capstone here.

3.7 Calculations

Comparison of the experimental and the theoretical accelerations:

$$a_{theo} = g \left(\frac{h}{L} \right)$$

$$\% error_a = \frac{|a_{theo} - a_{exp}|}{a_{theo}} \times 100$$

Comparison of the experimental and the theoretical gravitational accelerations:

$$\% error_g = \frac{|g_{theo} - g_{exp}|}{g_{theo}} \times 100$$